00-S-246d
August 8, 1967
SUPERSEDING
Int. Fed. Spec. 00-S-00246c (GSA-FSS)
August 23, 1965 and
Fed. Spec. 00-S-246b
March 10, 1964

FEDERAL SPECIFICATION

SAW, RECIPROCATING, PORTABLE, PNEUMATIC, WOODWORKING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type and size of portable pneumatic reciprocating woodworking saw for use in ripping, crosscutting, angle cutting and notching boards, timber, and logs.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-P-12829 - Packaging of Pneumatic Hand Tools with Attachments and Accessories.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C., 20402)

3. REQUIREMENTS

3.1 Illustration. The illustration shown herein is descriptive and not restrictive and is not intended to preclude the purchase of saws which otherwise conform to this specification.

3.2 Material. The materials used shall be of the highest quality suitable for the purpose intended. A combination of dissimilar materials that will cause galvanic corrosion shall not be used unless the saw design precludes such corrosion taking place.

3.3 Construction.

3.3.1 Ruggedness. The saw shall be of rugged construction so as to withstand, without damage, treatment likely to be encountered under general service conditions.

3.3.2 Air pressure. The saw shall be reliable and effective when operated on compressed air lines at gage pressures between 80 and 100 pounds per square inch.

3.3.3 Adjustment. The saw shall be self-contained, readily accessible for adjustment or replacement of parts, and of convenient size and shape for easy operation under service conditions.

3.3.4 Lubricant. Parts which require lubrication shall be properly enclosed so as to prevent entrance of foreign particles and leakage of lubricant.

3.3.5 Hardness. Working parts exposed to wear shall be of sufficient hardness to withstand the service required.

3.3.6 Finish. Parts shall be properly finished by machining or grinding to dimensions within limits established by good commercial practice. Exterior surfaces shall be suitably finished to resist corrosion.

3.3.7 Interchangeability. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances, in order that such parts may be completely interchangeable, and may be replaced or adjusted without requiring modification.

3.3.8 Air inlet connection. The air inlet connection shall consist of a well-secured 3/4-inch iron-pipe-size fitting threaded with female threads conforming to H28, fitted into rear handle.

3.4 Strainer. The saw shall be fitted with an air strainer or screen which shall be effective in retaining solid particles in the compressed air supply. The strainer shall be of such design that it can be easily removed for cleaning.

3.5 Marking for identification. Saws shall be permanently marked or provided with a suitable nameplate showing the manufacturer's model, type, or size designation, together with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.6 Operating temperatures. The saw shall be operable at temperatures ranging from minus 25°F to plus 125°F.

3.7 Handle.

3.7.1 Front handle. Each saw shall have an adjustable front handle of the closed type. The means by which this handle is adjusted shall be such that the handle will not become loose while the saw is in service.

3.7.2 Rear handle. The throttle sleeve shall be designed to form a rear handle.

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3.8 Throttle. A nonleaking, manual throttle shall be provided for admitting and shutting off the supply of compressed air. The throttle lever shall be of the rotating-sleeve style by means of which the air supply may be controlled and easily regulated. It shall be designed to stop the saw automatically when released.

3.9 Saw blades. Saw blades shall be of the twin interlocked type, fully heat-treated to a Rockwell hardness of not less than 42 nor more than 52 on the C scale. Saw blades shall be hard chromeplated and shall be free of projections that might interfere with operation in confined spaces. Each saw blade shall be approximately 26 inches long, 3 inches wide, and shall be properly set and sharpened.

3.9.1. Saw-blade lugs shall be firmly attached to the saw blade, and of a shape and size suitable to fit into saw-blade lug-retaining slots in the piston.

3.10 Saws shall conform to table I and shall be similar to figure I.

TABLE I. Detail saw requirements

Weight (max.)	Length overall (approx.) blade included	Cutting capacity	No Load air consumption (at 90 p.s.i.)
Pounds	Inches	Inches	Cubic feet/min.
20	46	1 to 21	50 to 80

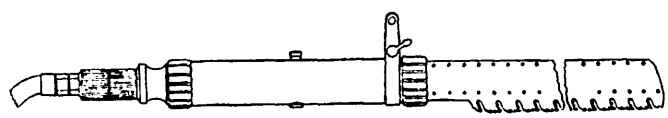


FIGURE 1. Pneumatic reciprocating saw.

3.11 Standard equipment. Each saw shall be furnished complete, consisting of the following:

1 Power unit, completely assembled, including rear control handle, and adjustable front handle.
1 Twin interlocked chromeplated saw blade assembly, 21-inch cutting capacity, sharpened and set.
1 Set of special tools (if required).
1 8-inch, single cut cant saw file.
1 Air-line oiler (see 3.12).
1 Carrying case (see 3.13).
1 Instruction book (see 3.14).

3.12 Air-line oiler. An air-line oiler of approximately 1/2-pint capacity shall be furnished with each saw. This oiler shall be adjusted so as to provide for an effective oil flow of not less than 2 ounces of oil per hour of saw operation.

3.13 Carrying case. Each saw shall be provided with either a metal or a wooden carrying case. This case shall be equipped with a suitable carrying handle and lid fastenings, be of sufficiently rugged construction to withstand, without injury, treatment likely to be encountered under general service conditions. Each case shall contain space therein for a completely assembled saw and the standard equipment specified in 3.11. The case shall be painted or suitably coated in accordance with the manufacturer's standard practice.

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3.14 Instruction book. The contractor shall furnish one copy of an instruction book, sealed in a waterproof envelope, with each saw supplied. When additional copies of the book are needed, the quantity and mailing addresses shall be specified in the contract or order.

3.15 Performance. The saw, when in the atmosphere or submerged in salt water, shall operate in all positions when cutting logs. The minimum cutting rate shall be 100 square inches per minute and the free air consumption shall be not more than 80 cubic feet per minute when cutting seasoned fir with 90-pounds-per-square-inch gage (p.s.i.g.) air pressure at the tool. In the atmosphere the saw shall crosscut, with not more than four blade sharpenings, 3,000 square feet of cutting area through logs having a diameter of not more than 21 inches.

3.16 Accessories or extra equipment. When additional accessories or equipment are required, they shall be as specified (see 3.11 and 6.1).

3.17 Workmanship. The saws shall be free from all imperfections which may effect its general appearance, function, or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

TABLE II. Sampling data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	One Complete Saw in Case	II	2.5	Defects per Hundred Units	4.3.1
Dimensional and weight examination	do.	II	2.5	do.	4.3.2
Testing (a) Operational	do.	II	1.5	Percent Defective	4.6.1 and 4.6.3
(b) Sawing	do.	-------	-------	-------	4.6.2
(c) Blade Hardness	One blade	S-1	1.5	Percent Defective	4.6.4

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table III

TABLE III. Classification of defects

Defects	
1	Material defective or not suitable for the purpose intended.
2	Saw not readily accessible for adjustment or replacement of parts; size and shape not convenient for ease of handling.
3	Lubricated parts not properly enclosed.
4	All parts not interchangeable.

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TABLE III. Classification of defects (Cont'd)

Defects	
5	Air inlet not correct size; not well secured; threads non-conforming.
6	Air strainer not easily removable or missing.
7	Front handle not closed type; not adjustable; adjusting devices does not prevent losing proper tightness or adjustment.
8	Throttle sleeve not properly designed for rear handle.
9	Throttle defective, not rotating sleeve type.
10	Saw blade defective, not interlocking type; not chrome plated; not free of projections; not correct length or width; not properly set and sharpened.
11	Lug not firmly attached to saw blade; not suitable size and shape.
12	Saw not in conformance with table I.
13	Component parts missing, defective, not as specified; power unit not completely assembled.
14	Air-line oiler not correct capacity; does not provide required oil flow.
15	Carrying case not durable constructed; not ample capacity to house saw and accessories.
16	Parts not properly finished.
17	Marking not as specified.
18	Instruction books missing; not in sealed waterproof envelopes.

4.3.2 Dimensional and weight examination. Each sample unit shall be examined for any nonconformance with dimensional and weight requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.6.1 and 4.6.3.

4.5 Inspection of preparation for delivery. Preparation for delivery inspection shall be as specified in MIL-P-12829.

4.6 Test procedures.

4.6.1 Operational test. Each sample saw shall be subjected to tests under practical service conditions to determine the ease of operation, general serviceability, suitability, and durability of the several parts. Air consumption during this test shall be determined with either a liquid displacement meter or a flow meter of common type. Air pressure shall be 90 p.s.i. at the saw air inlet. Failure of the saw, or any parts thereof, to operate properly shall be cause for rejection.

4.6.2 Sawing Test.

4.6.2.1 In atmosphere. One saw from each lot shall be tested to determine compliance with 3.15. Failure of the saw to cut as specified shall be cause for rejection.

4.6.2.2 Submerged. One saw shall make 10 complete cuts through an 18-inch diameter log submerged in salt water (salination value same as sea water). The 10 cuts shall be made in succession, making three crosscuts with the saw right side up sawing downward, three crosscuts with the saw upside down sawing upward, and four rip cuts for a distance of 3 feet. The saw shall be completely submerged in the salt water during this test. Failure of the saw to complete the test or malfunctioning, damage to, or permanent deformation of components shall be cause for rejection.

4.6.2.2.1 Corrosion resistance inspection. After successful completion of the tests in 4.6.2.2, the saw shall be placed in its case and stored for a minimum of 48 hours. At the end of the 48 hour period, the saw shall be completely disassembled and examined to determine compliance with 3.2.

4.6.3 Operating temperature test. To verify compliance with 3.6, the saw, with air-line oiler attached, shall be placed in a chamber, maintained at minus 25°F. and 100 per cent relative humidity, until the oil has reached a constant temperature of minus 25°F. The saw shall then be removed and immediately placed in service, sawing logs, for a minimum of fifteen (15) minutes. Failure of the saw to operate, or evidence of damage, shall be cause for rejection. After satisfactory completion of the above test, the same procedure shall be followed using a chamber temperature of plus 125° F.

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4.6.4 Hardness. Tests shall be conducted to determine compliance with 3.9 prior to chrome plating the saw blades.

4.7 Certification of tests. When the tests specified in 4.6.2 have previously been performed on the saw being offered at the time of Invitation for Bids, the manufacturer may furnish certified proof that the saws being offered are identical to those previously tested and approved. Testing in accordance with 4.6.2 will therefore not be necessary.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Saws shall be protected in accordance with level A of MIL-P-12829.

5.1.2 Level C. Level C preservation and packaging shall be in accordance with standard commercial practice.

5.2 Packing.

5.2.1 Level A. Saws shall be protected in accordance with level A of MIL-P-12829. Unless otherwise specified (see 6.1), the maximum gross weight of shipping containers shall be 200 pounds.

5.2.2 Level B. Saws shall be protected in accordance with level B of MIL-P-12829. Unless otherwise specified (see 6.1), the maximum gross weight of shipping containers shall be 200 pounds.

5.2.3 Level C. The pneumatic hand tools, attachments, and accessories, preserved and packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

5.3.2 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Accessories or extra equipment when required (see 3.16).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(d) Marking requirements, if different (see 5.3).

MILITARY CUSTODIAN:
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Navy - SH

Review activity:
Navy - SH

Preparing activity:
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User activities:
Navy - MC, YD

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